



CHAPTER

Concentrate Tank

Buffers and recirculates the brine/reject stream, and doubles as the CIP reservoir

IN THIS CHAPTER

16.1	Overview and function	108
16.2	Key specifications	109
16.3	Operation	110
16.4	Maintenance	111
16.5	Troubleshooting	112

16.1 Overview and function

The concentrate tank (tag prefix **TC**) buffers the brine/reject stream as it recirculates through the SonixED™ stack(s) (chapter 10): rejected ions accumulate in this stream over multiple passes, increasing its conductivity, until it is drained for disposal or routed onward for brine valorization (chapter 3). This tank also does double duty as the **circulation reservoir during CIP cleaning** — HydroVolta's documented CIP procedure (chapter 12) explicitly circulates acid and hypochlorite cleaning solution through this same tank, which is why the concentrate tank's minimum level is a precondition for starting a CIP cycle.



Figure 16.1. Suggested figure: photograph of the concentrate tank, labeled with its connections (from-stack, to-stack recirculation, drain/valorization outlet, CIP circulation port).

NOTE

This tank's role as the CIP cleaning reservoir — what water/chemical is in it at each stage, the hazards involved, and how it gets cleaned — is significant and safety-relevant enough to warrant its own chapter: see chapter 17.

16.2 Key specifications

As with the diluate tank, no finalized standard-product capacity/ material specification exists — see chapter 15 for the two real tank orders on file, which are candidates but not confirmed as to role. Some documented configurations use **two** concentrate buffer tanks in series (tagged **BTC1/BTC2**) rather than one.

- **Level:** TC_Level1 (main tank); BTC1/BTC2 level alarms exist in some configurations.
- **Conductivity:** monitored via the “Concentrate Sensor Island” (candidate: KROHNE SMARTPAT COND, chapter 13).
- **Flow:** concentrate-stream flow instrumentation confirmed in the register lists reviewed (chapter 13); a documented alarm exists for concentrate flow below **500 l/hr** (appendix A).

16.3 Operation

Concentrate recirculates through this tank during **Production** mode, increasing in conductivity pass over pass; during **CIP** mode (chapter 12), the same tank instead circulates cleaning chemical. Do not start a CIP cycle without first confirming the concentrate tank's level and that the CIP injection nozzle is submerged (chapter 12).

16.4 Maintenance

TODO: no tank cleaning/inspection schedule was found. Because this tank carries the concentrated reject stream (and periodically, cleaning acid/ hypochlorite), scaling and chemical residue are more likely here than in the diluate tank — inspect and descale per chapter 7 once an interval is confirmed, and always flush with fresh feed water after a CIP cycle per the procedure in chapter 12.

16.5 Troubleshooting

HL_Alarm_TC/LL_Alarm_TC cover the main tank (plus equivalent alarms for BTC1/BTC2 where fitted); a documented alarm fires below 500 l/hr concentrate flow. On any level or flow alarm, check the relevant sensor per section 13.4 before assuming a process fault. If a CIP cycle fails to reach its target pH, verify the concentrate tank's level and injection nozzle first, per chapter 12.